

# Arduino-Based Flight Simulator Using Joystick

A hands-on engineering project demonstrating real-time aircraft control using an Arduino Uno and a 2-axis joystick module.

ENGINEERING PROJECT

EMBEDDED SYSTEMS



# What Is a Flight Simulator?

A flight simulator replicates the experience of flying an aircraft in a controlled environment – used for pilot training, research, and entertainment.

## Why It Matters

Simulators reduce training costs and risk while allowing safe practice of emergency scenarios.

## Why Arduino?

Arduino offers an affordable, open-source platform for building custom cockpit hardware and interfacing with simulation software.

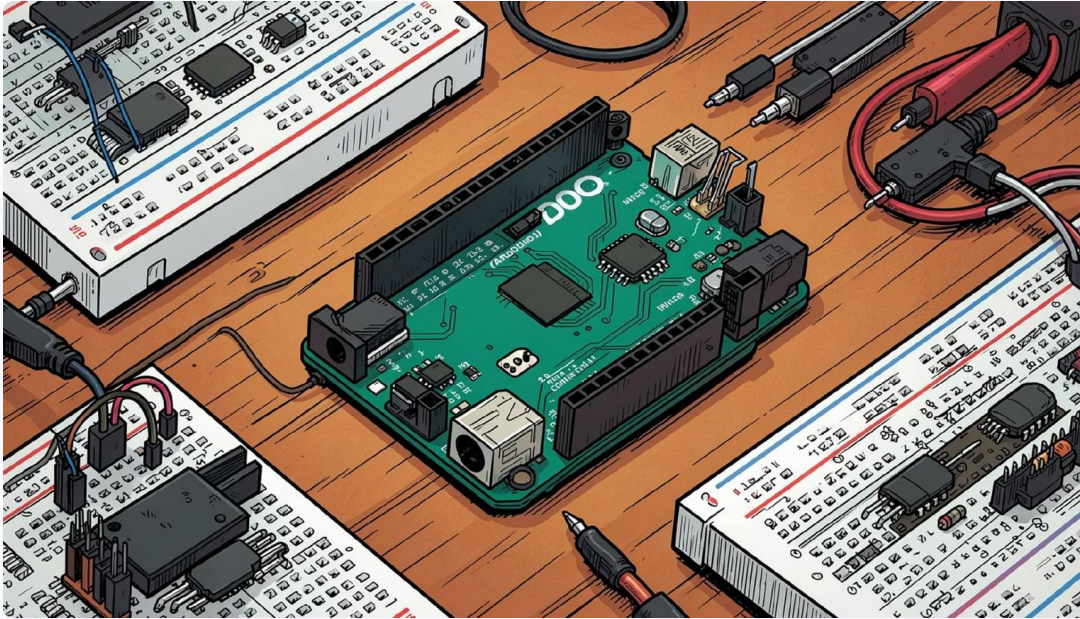
## DIY Approach

Students can replicate professional control systems at a fraction of the cost, gaining real embedded systems experience.





# Project Objectives



## What We Set Out to Achieve

- **Build a low-cost flight controller** using readily available components.
- **Learn Arduino programming** to read analog sensor inputs.
- **Interface hardware with simulation software** via USB serial communication.
- **Develop hands-on embedded systems skills** applicable to real-world engineering.

# Components Required



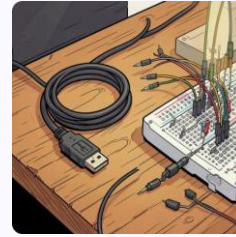
## Arduino Uno

The brain of the system – reads joystick inputs and sends data to the PC.



## 2-Axis Joystick Module

Provides analog X and Y axis readings plus a push-button switch.



## USB Cable & Wires

Connects the Arduino to the computer and links components on the breadboard.



## Computer / Laptop

Runs the flight simulator software and receives serial data from the Arduino.

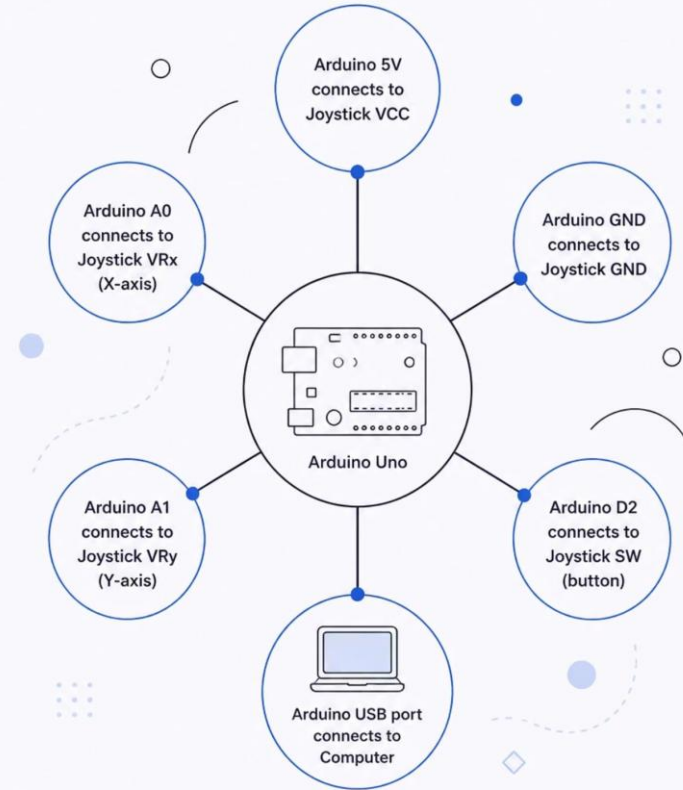
# Hardware & Circuit Connections

## Arduino Uno

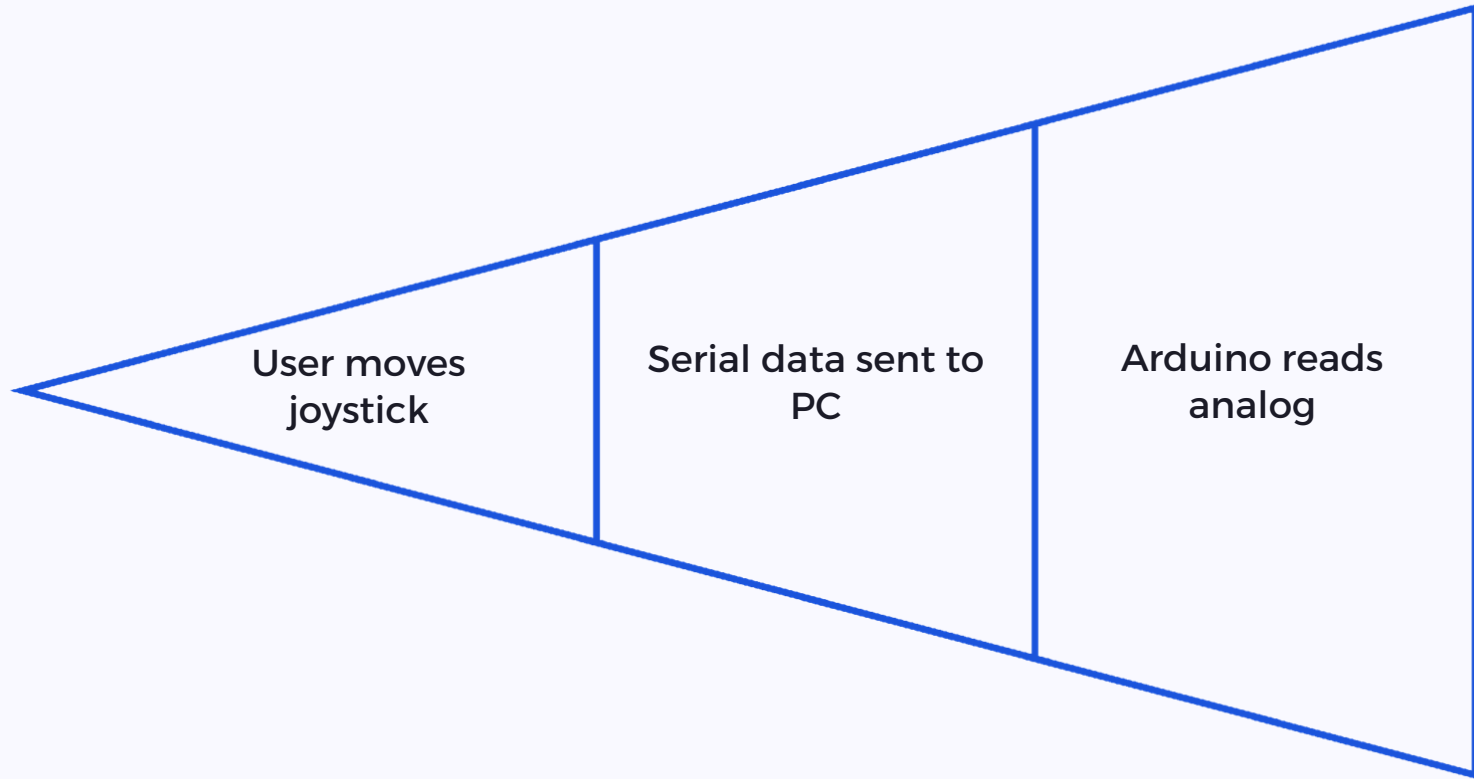
Based on the ATmega328P, the Uno features 14 digital I/O pins, 6 analog inputs, and a 16 MHz clock – ideal for reading joystick signals.

### Joystick Module Pins

- **VCC** → 5V power supply
- **GND** → Ground
- **VRx** → Analog pin A0 (X-axis)
- **VRy** → Analog pin A1 (Y-axis)
- **SW** → Digital pin D2 (button)

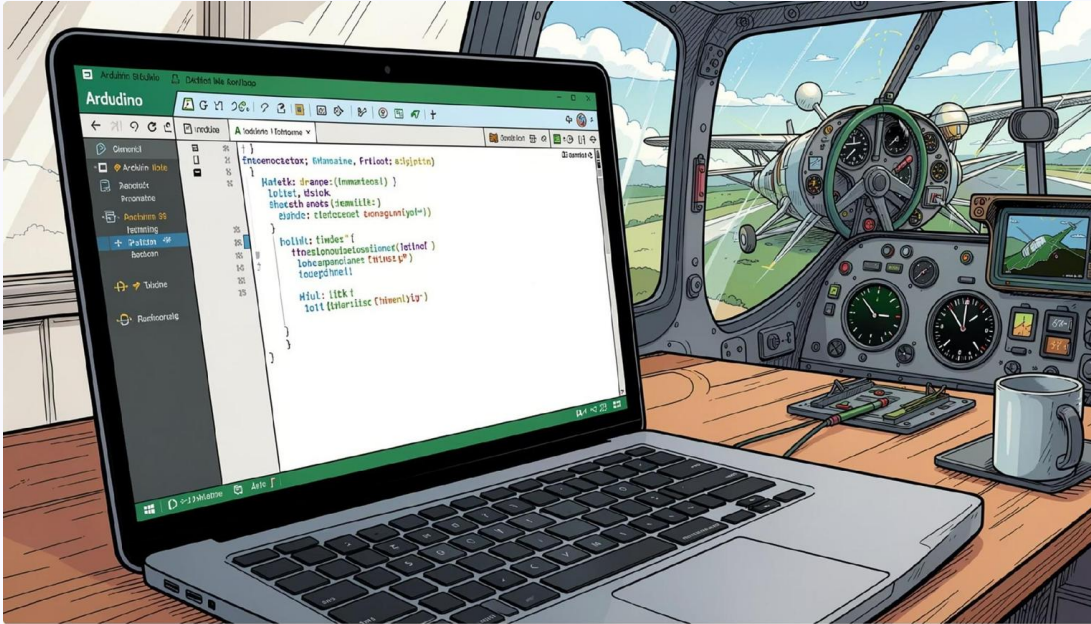


## Working Principle



The joystick outputs analog voltages that the Arduino converts to digital values. These values are mapped to aircraft control commands and transmitted over USB in real time, creating a responsive flight control experience.

# Software & Program Flow



## Software Stack

### Arduino IDE

Write, compile, and upload code to the Arduino board.

### Joystick Library

Maps analog inputs to HID joystick commands (optional).

### FlightGear / X-Plane

Open-source simulators that accept serial or joystick input.

## Program Flow

1

Read analog values from pins A0 and A1

2

Map values from 0-1023 to control range

3

Send serial data over USB to PC

4

Simulator updates aircraft pitch and roll in real time





# Live Demonstration

## ✈ Pitch Control

Pushing the joystick forward tilts the aircraft nose down; pulling back raises it./...//

1. ↔ Roll Control
2. Moving left or right banks the aircraft, simulating aileron input.

## ⚡ Real-Time Response

Changes are reflected in the simulator with minimal latency via USB serial.



# Advantages & Limitations

## ✓ Advantages

- **Low cost** – built with components under \$20
- **Easy to assemble** – minimal soldering required
- **Educational value** – covers electronics, coding, and aviation
- **Real-time control** – responsive aircraft movement
- **Expandable** – add throttles, switches, and pedals

## ⚠ Limitations

- **Basic controls only** – pitch and roll, no full 6-axis
- **No force feedback** – joystick lacks resistance cues
- **Limited precision** – analog noise affects accuracy
- **Software dependency** – requires a PC and simulator

# Future Scope & Conclusion



## Throttle & Rudder

Add potentiometer-based throttle and rudder pedals for full 3-axis control.



## Cockpit Instruments

Drive LCD displays and LED indicators to replicate real cockpit gauges.



## VR Integration

Combine with a VR headset for an immersive, head-tracked flight experience.



**Conclusion:** This project demonstrates how Arduino and a simple joystick can create an affordable, interactive flight simulator – bridging embedded systems, software, and aviation in a single hands-on build.

# Thank You

Questions & Answers

